

Table 4. Choice random-parameter logit panel model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health attributes			
NOSE*LNDAYS	0.4522**	0.9675	2.12
Estimated SD	—		
EYE*LNDAYS	0.1653**	0.7995	2.07
Estimated SD	0.0001		0.00
FLUTTER*LNDAYS	−0.1851***	0.5944	−2.17
Estimated SD	0.0134		0.02
BREATH*LNDAYS	−0.0786	0.6568	−0.94
Estimated SD	0.0081		0.01
ACHE*LNDAYS	0.1101	0.7672	1.19
Estimated SD	0.0074		0.01
SWELL*LNDAYS	−0.1916*	0.5906	−1.99
Estimated SD	0.1016		0.29
PAIN*LNDAYS	−0.2723***	0.5434	−3.23
Estimated SD	0.0273		0.04
NOLIM	0.5078*	1.0000	1.78
Estimated SD	—		
SOMELIM	0.4864***	0.9875	4.87
Estimated SD	0.3133**		2.10
NOSOC	0.5998***	1.0538	6.06
Estimated SD	0.4752***		3.57
ATHOME	0.0038	0.7050	0.04
Estimated SD	0.5444***		3.82
NEEDHELP	−0.3971***	0.4704	−3.30
Estimated SD	0.5104***		2.59
INHOSP	−1.2007***	0.0000	−6.22
Estimated SD	1.4752***		10.01
Utility of money function (*SQTCOST)			
CONSTANT	−0.1967***		−9.35
SCORE	−0.0000276***		−3.69
AGE	0.000323		1.16
EDUCATION	0.000716***		8.81
SYMPTOMATIC	0.008012*		1.72
VIMPORTANT	−0.011477***		−3.33
Current health function			
ASTHMA*FREQHOME	−0.0953***		−4.39
BRONCHITIS	−0.7115***		−3.21
LUNGINFECTION	−0.2569***		−4.42
HEARTDISEASE*FREQHOME	−0.1335***		−3.98
AGESQUARED	0.0000508***		2.56
INCOME	0.002608***		7.46
PAINRATING	0.003662***		3.72
VIMPORTANT	0.0986***		4.54
FREQILL	−0.1030***		−6.94
FREQHOME	0.2059***		7.14
Alternative-specific constants			
CH_IC	−1.2888***		−4.29
CH_A	0.2423		1.24
Likelihood ratio Chi-squared		557***	
McFadden <i>R</i> -squared		0.093	
Percent correctly predicted		39.3	
Number of observations		2752	

* Significant at the 10% level or better; ** significant at the 5% level or better; *** significant at the 1% level or better.

when interacted with FREQHOME. Thus, ASTHMA and HEARTDISEASE are associated with lower than average health only when they are severe enough to require staying home. Of the four conditions, chronic bronchitis is by far the most serious, with disutility 2.5 times greater than the next most serious condition, while asthma is the least serious.

Age-squared is also positive and highly significant. Generally we expect that older people experience lower levels of health than younger people. It is worth noting, however, that we are modelling subjective health utility, not health *per se*. This result may be interpreted as saying that older people obtain higher levels of utility from the health they have. This is consistent with the positive sign on AGE in the utility of money function, which says that older people value money less than health compared with younger people.

Recall that the experimental design required that cost be ordered from zero to high values among the initial condition IC and alternatives A and B, while activity restrictions are ordered inversely. Thus, some subjects may have tended to choose A simply because it had better health than IC and lower cost than B. CH_A is significant in models where money and current health functions are treated as constants, but insignificant when these functions are included. This result means that including utility of money and current health functions eliminates the middle alternative bias. However, the significant negative coefficient for CH_IC indicates that subjects chose IC relative to A and B less often than the explanatory variables alone would predict.

JOINT GRADED-PAIR, DISCRETE-CHOICE ESTIMATES

Our goal is to obtain estimates suitable for use in a wide range of benefit–cost applications. Recall that the graded-pair format is designed to obtain information on marginal trade-offs among health attributes and cost because there is no unambiguous baseline reference point and no health conditions with zero cost. On the other hand, the discrete-choice format can obtain values for changes in health relative to an identified reference point, but reduces the salience of the symptom attribute because symptom levels do not vary within a given choice set. Consequently, because each SP format has strengths as well as limitations, we pool the data

and estimate a joint model to capture all the available information on subjects' preferences in one set of parameters. Estimating overall preferences in a joint model thus provides a more complete picture of subjects' utility functions for calculating WTP than does either format individually.

Although it would be interesting to estimate a joint panel model, such estimates are beyond the scope of the present study. Instead, we estimate a pooled model using the same specifications as the panel models discussed above, but do not control for intra-subject correlations.¹¹ We then compare the results with the individual panel models.

From Equation (1),

$$U_{st}^i = \lambda_f \cdot V^i(X_{st}, Z^i, P_{st}; \beta, \delta_s^i) + e_{st}^i, \quad (14)$$

where now s indexes an attribute profile in either a graded-pair or choice repetition. We include a choice utility scale function to control for different variances in the two data sets, so $\lambda_f = \lambda_c$ for the choice format and $\lambda_f = 1$ for the graded-pair format.^{mm} The utility parameters β are constrained to be the same for both elicitation formats. However, the utility of money function parameters are estimated separately. Thus,

$$\delta_f^j(Z^i) = \left[\gamma_{f0} + \sum_m (\gamma_{fm} \cdot Z_m^i) \right], \quad (15)$$

where again f indexes the graded-pair and choice formats, respectively. We include the same utility of money variables used in previous models. The joint model also incorporates the current health condition variables used previously, so that WTP values can be estimated relative to different individual current health reference points. Information on current health utility comes only from the choice format. Finally, we combine SOMELIM and NO-SOC into a single category. Previous results indicated that subjects discriminated poorly between these two mild restriction categories.

Figure 3 compares the joint estimates with the discrete-choice and graded-pair panel models. The joint estimates fall between the two panel estimates in nearly every case, although they are clearly not a simple average of the two. The joint estimates track the graded-pair point estimates somewhat more closely than the discrete-choice estimates for the more severe symptoms. Conversely, the joint estimates track the discrete-choice estimates more closely for the activity categories. Thus, the joint estimates have the desired property of giving more weight to the format with the smaller variance for each parameter, i.e. the more precise estimate.

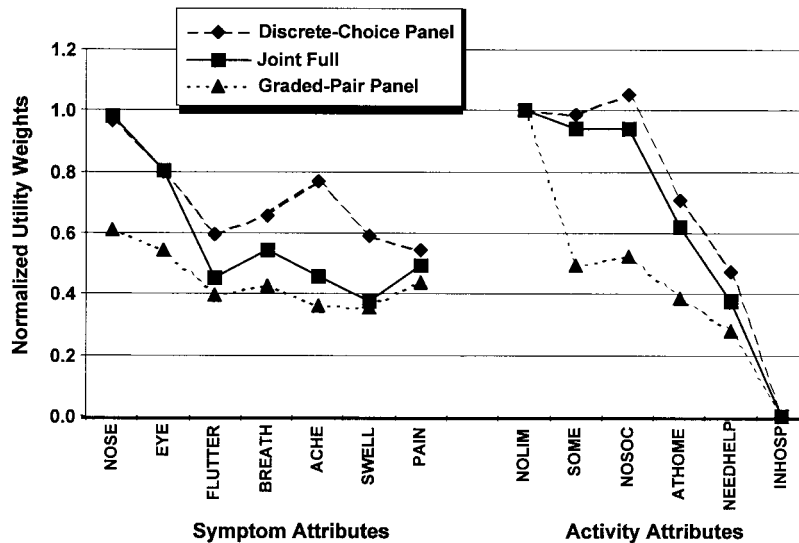


Figure 3. Comparison of joint and individual-panel models

Table 5 reports estimates for the joint model. As in previous models, we present both nominal and normalized coefficients. The pattern of normalized symptom utility weights reflects more variability than either the graded-pair or choice utility weights separately, as also seen in Figure 3. The coefficients for NOSE and EYE are significantly different from the other symptom variables. In addition, SWELL is also significantly different from the other variables. In addition, all variables are significantly different from the mean symptom effect, and only BREATH is significantly different than the mean effect at the 10% level. Thus, the symptom levels in the joint model exhibit substantial variation and more significance than either of the two separate models.

As in the choice model, only ATHOME is statistically insignificant relative to the mean activity effect. However, *t*-tests show that ATHOME is significantly different than the milder category of SOMELIM/NOSOC and activity levels decrease monotonically as daily activity restrictions become more severe. While NOLIM and SOMELIM/NOSOC are significantly different than the mean effect, they are statistically equivalent to each other. Finally, as we observed in Figure 3, the variation in activity levels reflects the choice model more closely than the graded-pair model.

Utility of money functions are estimated separately for the graded-pair and discrete-choice data

to account for systematic differences between the two formats. The implicit marginal utility of money is much smaller in the graded-pair data than the discrete-choice data. There are two possible reasons for the large difference in marginal effects. First, the nature of the discrete-choice task induces subjects to focus on the attribute levels themselves, rather than on differences among the attribute levels. Most people solve the choice valuation problem by eliminating undesirable alternatives based on one or two unattractive attribute levels. Consequently, the task itself simply may cause subjects' to pay more attention to the cost attribute in evaluating alternatives. Second, because symptoms were held constant across two of the three alternatives, subjects did not have to pay as much attention to this attribute as they did in the pairs task. Therefore, they may have had more cognitive energy to focus on the other attributes, including cost. The other utility of money coefficients exhibit similar patterns as in the previous two models.

In the joint model, the relative scale function fixes differences in variance in the two data sets. We use the same scale variables as before, but their interpretation is different in this model. The relative scale function indicates how the variance or noise varies systematically in the choice data set relative to the pairs data set. A positive coefficient indicates a source of lower variance in the choice data. Overall, this function is positive,

Table 5. Joint model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health Attributes			
NOSE*LNDAYS	0.2572***	0.9829	5.52
EYE*LNDAYS	0.1405***	0.8035	7.04
FLUTTER*LNDAYS	−0.0871***	0.4535	−4.59
BREATH*LNDAYS	−0.0294*	0.5422	−1.69
ACHE*LNDAYS	−0.0831***	0.4596	−4.08
SWELL*LNDAYS	−0.1372***	0.3764	−7.91
PAIN*LNDAYS	−0.0609***	0.4938	−3.05
NOLIM	0.2683***	1.0000	4.47
SOMELIM/NOSOC	0.2299***	0.9410	9.39
ATHOME	0.0225	0.6220	0.84
NEEDHELP	−0.1387***	0.3741	−5.36
INHOSP	−0.3820***	0.0000	−9.48
Utility of money function (*SQT COST)			
CH_CONSTANT	−0.1072***		−6.54
CH_SCORE	−0.0001819**		−2.17
CH_AGE	0.000297*		1.65
CH_EDUCATION	0.003968***		5.13
CH_SYMPOMATIC	0.003815		1.03
CH_VIMPORTANT	−0.004019**		−2.09
GP_CONSTANT	−0.007188		−0.40
GP_SCORE	−0.0002305**		−2.22
GP_AGE	0.000432**		2.44
GP_EDUCATION	−0.0005507		−0.53
GP_SYMPOMATIC	0.008877*		1.82
GP_VIMPORTANT	−0.004918**		−2.17
Current health condition			
CH_ASTHMA*FREQHOME	−0.0658***		−2.58
CH_BRONCHITIS	−0.3523**		−2.29
CH_LUNGINFECTION	−0.1679***		−2.53
CH_HEARTDISEASE*FREQHOME	−0.1621***		−2.50
CH_AGESQUARED	0.0000267*		1.62
CH_INCOME	0.001624***		4.39
CH_PAINRATING	0.002615***		2.46
CH_VIMPORTANT	0.0466***		3.27
CH_FREQUILL	−0.0496***		−2.85
CH_FREQHOME	0.1124***		3.44
Relative scale function			
CH_HISORE	0.9345***		5.03
CH_HIAGE	−0.8042***		−3.69
CH_VIMPORTANT	0.0718		1.20
CH_NOPAIDLEAVE	−0.1327		−0.77
CH_HIINFORMATION	−0.3505*		−1.80
CH_FAILCHECKS	0.1819		0.89
Alternative-specific constants			
CH_A	−0.5327***		−3.28
CH_B	0.2809**		2.14
GP_ALPHA1	−1.1330		
GP_ALPHA2	−0.1635		
GP_ALPHA3	0.6159		
Likelihood ratio Chi-squared		827***	
McFadden <i>R</i> -squared		0.063	
Graded-pair percent correctly predicted		42.1	
Choice percent correctly predicted		47.5	
Number of observations		5504	

* Significant at 10% level; ** significant at 5% level; *** significant at 1% level.